

## ISELECT2 – IS ELECTIVE 2 (HCI)

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### **A Critique Paper for eGov PH Super App under Department of Information and Communications Technology (DICT)**

#### **I. OVERVIEW**

##### **A. Website Overview**

The Department of Information and Communications Technology responsible for the operation of government data centers, the National Broadband Plan, public private partnerships in ICT infrastructure development, regulations of telecommunications services and products, telecommunications standards development and promotions of digital literacy programs [1]. The eGov PH Super App, developed by the Philippines' Department of Information and Communications Technology, consolidates online government services into a single app, enabling seamless transactions and single sign-on access. It supports the Philippine National Digital ID project and facilitates seamless payments through eGovPay. The app also allows citizens to secure vital documents from their homes, reducing bureaucratic red tape and making government services more efficient and accessible. The DICT in collaboration with other government agencies initially introduced the eGov Super App on December 2, 2022. The initiative involves developing, implementing, and integrating several components, such as E-Local Government Units (ELGU), E-Government Application (EGovApp), EGovpay, E-Travel, and E-Cloud, among others [2].

The DICT website has a professional and effective design, with a major focus on the "ANG BAYANG DIGITAL, BAGONG PILIPINAS 2024" concept and core services. The website includes thorough information on essential projects like Free Public Wi-Fi and the National Broadband Plan, which help the government achieve its digital transformation goals.

The Programs area is comprehensive and well-organized, allowing visitors to easily navigate programs such as Tech4ED, which encourages rural technology access. The website stays flexible on mobile devices, and the Contact Us section has various contact methods, making it simple for consumers to get in touch. Overall, the website reflects DICT's dedication to improving the country's ICT landscape and digital inclusiveness.

##### **B. Link**

<https://dict.gov.ph/>

##### **C. System User**

The Department of Information and Communications Technology (DICT) supports a diverse set of system users, including:

**Government Agencies:** DICT's technologies are used by a variety of government offices and institutions to assist e-government projects, data management, cybersecurity standards, and infrastructure development.

**Private Sector Companies:** DICT provides regulatory compliance, digital infrastructure, and policy support to businesses, particularly those in telecommunications, software development, and ICT-enabled services (ICT-ES).

**Educational Institutions:** Schools, universities, and training centers collaborate with DICT to use ICT resources for digital learning platforms, research, and skills development.

**Law Enforcement Agencies:** The Philippine National Police (PNP) and other law enforcement agencies employ DICT systems for cybersecurity, digital forensics, and cybercrime investigations.

**Citizens and Communities:** DICT projects like as Free Public Wi-Fi and Tech4ED assist individuals, particularly those in rural regions, by providing access to the internet, e-learning, and digital services.

**ICT Professionals:** IT specialists, cybersecurity experts, and technology consultants who collaborate with DICT to improve digital services, safeguard infrastructures, and drive the country's technical advancement.

## II. Methodology

### A. Research Design

A quantitative questionnaire has been adopted and utilized by the researchers as a survey instrument. The researchers used a quantitative questionnaire as their survey tool to collect and analyze numerical data in an organized way. This method makes it easier to get consistent responses from the participants. It also helped them identify patterns and trends in the data, resulting in more credible and broadly applicable conclusions. Using numbers also allowed the researchers to compare and connect the data, which improved the accuracy and validity of their conclusions.

### B. Research Locale



**Figure 2.** Research Locale

This study was conducted at Davao del Norte State College in Panabo City, Davao del Norte, Philippines. The respondents were surveyed using Google Forms researchers selected this location for the implementation of the study as it would provide them with a sufficient amount of information regarding

the Department of Information and Communication Technology (DICT) System. The study was conducted in the First semester of the academic year 2024-2025.

### **C. Participants of the Study**

The study's participants are Davao Del Norte State College students enrolled in programs offered by the Department of Information and Communication Technology (DICT). They were chosen based on demographic criteria such as age and gender to ensure a broad and representative sample. The study intends to involve a diverse group of respondents of various ages and genders in order to provide an accurate representation of the DICT system's impact. Statistical analyses will be performed to describe the individuals' sociodemographic features and origins.

### **D. Sampling Techniques**

The researchers must collect data from important people in order to answer the study questions. A successful sampling technique aims to guarantee that the conclusions gathered from the sample are applicable to a larger population. This approach enables a thorough analysis of the research inquiries and ensures that all important information is accounted for [6]. Having clearly specified participants improves the validity and reliability of the findings for this study, thus the researchers decided to use convenience sampling technique to measure the sample size for this study.

Convenience sampling is a simple and easy way to get information compared to other sampling methods. This type of sampling is not at all based on probability and is commonly utilized in research studies. People whom the researcher can easily get to contact and find are selected applying this sample technique. Convenience sampling is often used because alternative techniques of sampling are hard or impossible to use due to time, money, or other constraints. Of course, it could be a rapid and simple means of data gathering but may also introduce biases and limitations that compromise the reliability and practicality of results [7].

### **E. Statistical Treatment**

The data from the questionnaire were compiled into tables and then given various statistical treatments.

**1. Percentage and Frequency Distribution.** This method was used to classify the respondents according to their sex and age group. In addition to percentages, the ranking was utilized to describe numerical data.

**2. Weighted Mean.** This method was used to categorize the distinct levels of Usability Evaluation, User Interface Design, Performance Evaluation, and Continent Accessibility Evaluation.

### **F. Data Collection Procedure**

The first step before moving on to the real testing is to ask questions. The second stage is to get informed consent from our target respondents. After the respondents accepted informed consent, the researcher sent them an online survey questionnaire. The selected respondents were chosen online at Davao Del Norte State College. They were proposed based on the questions provided via the Google Form questionnaire. A survey conducted online is the systematic collection of data from a target audience, defined by the invitation of respondents and the completion of a questionnaire through the Internet [5]. Following data collection, the data were tallied, and tabulated, and statistical treatments were applied.

#### D. Research Instrument

The research instrument utilized in the study was a researcher-created questionnaire, a Likert scale designed to collect essential data from the students. This questionnaire was crafted based on the researcher's review of relevant readings, previous studies, as well as published literature related to the study. During the development of this instrument, careful attention was paid to the requirements for creating an effective data-collection tool. For example, the wording of statements describing specific situations was adjusted to align with the respondents' level of knowledge and comprehension. This strategic approach ensured that the questionnaire was well-suited to gather the necessary information from the participants in a manner that was both comprehensive and accessible. Questionnaires are widely employed in quantitative marketing and social research. A questionnaire is a collection of questions asked to persons to collect statistically meaningful information about a specific topic. A questionnaire is a list of inquiries made to people to gather data on a statistically significant. When properly constructed and responsibly administered, questionnaires become a vital instrument by which statements can be made about specific groups or people or entire populations [3].

#### E. Ethical Consideration

The main concern of the study is college students. The researchers ensure the students' safety and confidentiality, and they follow ethical standards. These are the following standards: consent, respect for the students, justice, and confidentiality. The participants of the study were asked for their permission, approval, and time to answer the questionnaire through Google Form which is already written in the link and was sent via Messenger and Facebook. A survey involves the important use of Google, as well as lengthy processes to generate the necessary information [4]. Consent is also one way of showing respect to the participants. Through consent, the participants were able to know the objectives and purpose of the study. The consents were given through messenger and Facebook, and after they answered the consent, they proceeded to answer the questionnaire. For the participants' protection, anonymity, and confidentiality of any of the responses from them were observed. The researchers were also responsible for the participants' identities and kept the notes, transcripts, and other materials used from the conducted survey. It is important to give credit or acknowledge the participant's contributions as they are part of the research's success. This is to give justice to their participation in the said study.

### III. RESULTS AND DISCUSSION

The study outcomes are discussed in this chapter. The presentation of data in this research study are arranged in the following sequence: Demographic profile of the participants in terms of gender and age, the usability of DICT system, the user interface design of the system, and lastly the performance of the said system.

#### *Demographic Profile of the participants*

There are 30 total respondents to this study composed of 20 females and 10 males. Participants in the age group of 18-23 years old has the highest percentage, meaning most of the respondents are aged between 18 to 23 years old. Table 1 shows these findings.

Table 1. Demographic Characteristics of the Participants

| Demographic Profile | Frequency<br>n=30 | Percentage |
|---------------------|-------------------|------------|
| Sex                 |                   |            |

|                  |    |      |
|------------------|----|------|
| Female           | 20 | 66.7 |
| Male             | 10 | 33.3 |
| <b>Age Group</b> |    |      |
| 18-23            | 24 | 80   |
| 24-29            | 5  | 16.7 |
| 30 above         | 1  | 3.3  |

#### *Usability Evaluation of DICT system*

| Indicators       | Mean | Standard Deviation | Descriptive Equivalent |
|------------------|------|--------------------|------------------------|
| Learnability     | 4.7  | 0.52               | Strongly Agree         |
| Efficiency       | 4.7  | 0.52               | Strongly Agree         |
| Memorability     | 4.73 | 0.51               | Strongly Agree         |
| Error Prevention | 4.8  | 0.4                | Strongly Agree         |
| Satisfaction     | 4.73 | 0.51               | Strongly Agree         |
| <b>Overall</b>   | 4.73 | 0.49               | Strongly Agree         |

#### *User Interface Design*

| Indicators         | Mean | Standard Deviation | Descriptive Equivalent |
|--------------------|------|--------------------|------------------------|
| Visual Design      | 4.76 | 0.49               | Strongly Agree         |
| Consistency        | 4.8  | 0.4                | Strongly Agree         |
| Navigation         | 4.7  | 0.52               | Strongly Agree         |
| Responsiveness     | 4.6  | 0.6                | Strongly Agree         |
| Emotional Response | 4.6  | 0.6                | Strongly Agree         |
| <b>Overall</b>     | 4.69 | 0.52               | Strongly Agree         |

#### *Performance Evaluation*

| Indicators         | Mean | Standard Deviation | Descriptive Equivalent |
|--------------------|------|--------------------|------------------------|
| Load time          | 4.6  | 0.55               | Strongly Agree         |
| Responsiveness     | 4.6  | 0.55               | Strongly Agree         |
| Page Size          | 4.5  | 0.61               | Strongly Agree         |
| Server Performance | 4.5  | 0.61               | Strongly Agree         |
| Functionality      | 4.5  | 0.61               | Strongly Agree         |
| <b>Overall</b>     | 4.54 | 0.59               | Strongly Agree         |

#### *Content and Accessibility Evaluation*

| Indicators                     | Mean | Standard Deviation | Descriptive Equivalent |
|--------------------------------|------|--------------------|------------------------|
| Relevance of Information       | 4.4  | 0.55               | Agree                  |
| Easy Access Content            | 4.3  | 0.54               | Agree                  |
| Accessibility for Disabilities | 4.3  | 0.54               | Agree                  |
| Content Clarity                | 4.4  | 0.55               | Agree                  |
| Multimedia Use                 | 4.3  | 0.54               | Agree                  |
| <b>Overall</b>                 | 4.34 | 0.54               | Agree                  |

#### **IV. CONCLUSION**

The usability review of the Department of Information and Communications Technology (DICT) system, based on comments from Davao del Norte State College students, showed an overall satisfactory user experience. Participants evaluated the system to be user-friendly, with high learnability, ease of navigation, and task efficiency. The user interface was complimented for its simplicity, clear structure, and uniformity between parts, which enabled smooth interaction.

In terms of performance, the system was responsive and stable, however minor concerns such as occasional delays or sluggishness were reported. Content accessibility, notably the relevance and availability of information, fulfilled user expectations; nevertheless, certain areas, such as accommodating users with impairments and improving multimedia integration, were noted as requiring more improvements.

## V. RECOMMENDATION

The findings indicate the Department of Information and Communications Technology (DICT) is dedicated to developing ICT skills and connections through a variety of initiatives.

It is recommended that:

1. Students should engage in DICT programs and training to enhance their ICT skills and explore how technology can improve their learning experience.
2. Instructors are advised to integrate ICT tools into their instruction, promote DICT activities, and engage with DICT on curriculum creation to ensure alignment with industry standards.
3. Future researchers should concentrate on examining new ICT trends, working with DICT to gain access to resources and data, and investigating the role of ICT in social and economic development, particularly in underprivileged communities.

## VI. REFERENCES

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## VII. APPENDICES

### Questionnaire Link

[https://docs.google.com/forms/d/e/1FAIpQLSeFfl-83oRgLq2\\_NQ3Xc5zD2fry\\_8e4dVxJAndvY8bFHlfKzg/viewform?usp=sharing](https://docs.google.com/forms/d/e/1FAIpQLSeFfl-83oRgLq2_NQ3Xc5zD2fry_8e4dVxJAndvY8bFHlfKzg/viewform?usp=sharing)

### Responses Link

[https://1drv.ms/x/c/2a324a191c2e20b0/Ec\\_nNZRXACdGIhfgmwb1cr4BloCFkAXuA13qJ8ZX09ObFQ?e=inyPnX](https://1drv.ms/x/c/2a324a191c2e20b0/Ec_nNZRXACdGIhfgmwb1cr4BloCFkAXuA13qJ8ZX09ObFQ?e=inyPnX)